

WHEN MORE ROLLOUTS REMEMBER THE WRONG THING: BEST-OF-N AMPLIFIES STALE MEMORY IN RETRIEVAL-AUGMENTED WORLD MODELS

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ABSTRACT

Memory-augmented world models retrieve episodic transitions, trajectories, or cases to improve prediction under sparse data. Separately, Best-of-N inference samples many candidate futures and selects the one with the highest model proxy. This paper studies a narrow interaction between the two: stale or support-mismatched memory can create rare *memory-impostor* rollouts with high predicted value and low true value, and Best-of-N makes those rollouts more likely to be selected. We give a finite-N mechanism statement, propose diagnostics based on retrieval precision, staleness, counterfactual memory dropout, and proxy-true gap, and instantiate the mechanism in a controlled latent-regime world. In the main stale-memory condition, naive memory Best-of-N improves its proxy score while true return falls from -5.30 at $N = 1$ to -13.29 at $N = 64$; selected rollouts have 13.7% retrieval precision, 86.3% stale retrieval, and 27.8% hallucinated-rollout rate. A repair using recency retrieval, disagreement-aware memory gating, and dropout-risk penalties reaches -1.95 true return at the same N , narrowing but not eliminating the gap to an oracle retriever. The claim is intentionally scoped: this is a runnable diagnostic for an architecture-specific failure mode, not evidence that memory is generally harmful.

1 INTRODUCTION

World models let agents reason in imagined futures rather than only through environment interaction (Sutton, 1990; Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; Schrittwieser et al., 2020). A common inference pattern is to sample many candidate rollouts and select the one with the highest model score. This is attractive whenever the generator is cheap and the scoring proxy is believed to be reliable.

External memory gives a second source of test-time power. Episodic control, retrieval-augmented reinforcement learning, case-based reasoning, kNN language models, and retrieval-augmented generation all show that non-parametric memory can improve adaptation and factuality (Aamodt & Plaza, 1994; Pritzel et al., 2017; Wayne et al., 2018; Khandelwal et al., 2020; Guu et al., 2020; Lewis et al., 2020; Goyal et al., 2022). For world models, the hope is direct: retrieve relevant cases and make better local transition predictions.

This paper asks what happens when these two useful ideas are composed under support mismatch. If the retriever returns stale cases from a latent regime with the wrong transition sign, a memory-augmented model can assign high value to a candidate future that the real environment will not realize. Best-of-N then searches for that error. More samples need not make the selected rollout more grounded; they can make it a better impostor under the memory-biased proxy.

Our contributions are:

- a finite-N statement of memory-impostor amplification;
- diagnostics for selected-rollout retrieval precision, staleness, memory dropout variance, hallucinated-rollout rate, and the memory exploitation gap;

- a controlled synthetic world where latent dynamics, memory provenance, and true rollout utility are all known;
- a simple repair that combines recency-aware retrieval, disagreement-aware memory gating, and an uncertainty penalty.

2 RELATED WORK

World models and model-based RL. Model-based RL has long used learned dynamics for planning (Sutton, 1990). Modern world models learn latent dynamics from high-dimensional observations and use imagined trajectories for control (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020). Probabilistic and offline model-based methods such as PETS, MBPO, MOREL, and MOPO explicitly address model bias and uncertainty (Chua et al., 2018; Janner et al., 2019; Kidambi et al., 2020; Yu et al., 2020). Our setting is not a new planner; it isolates a failure source that arises when the local dynamics model is partly retrieved from an external case memory.

Episodic, retrieval, and case-based memory. Case-based reasoning formalized retrieve/reuse/revise/retain cycles (Aamodt & Plaza, 1994). Neural episodic control and MERLIN showed that fast memory can help agents adapt (Pritzel et al., 2017; Wayne et al., 2018), while retrieval-augmented RL and model-based episodic memory bring retrieved experience into control and model-based settings (Goyal et al., 2022; Le et al., 2021). Retrieval-augmented language models and kNN-LMs show a closely related non-parametric interpolation pattern (Khandelwal et al., 2020; Guu et al., 2020; Lewis et al., 2020; Borgeaud et al., 2022; Wu et al., 2022). These works motivate memory augmentation; our contribution is to show how inference-time selection can amplify a memory error that average prediction metrics may understate.

Test-time selection and proxy optimization. Self-consistency, verifier ranking, Tree of Thoughts, and test-time compute scaling all improve outputs by sampling or searching multiple candidates (Wang et al., 2023; Cobbe et al., 2021; Yao et al., 2023; Snell et al., 2024). Reward-model overoptimization shows that selecting against an imperfect proxy can degrade true utility (Gao et al., 2023). Our finite-N argument is simple in that tradition; the novelty is the memory-specific source of proxy/true inversion and the corresponding retrieval diagnostics.

3 MECHANISM

Let a candidate rollout be grounded with probability $1 - p$ and a memory-impostor with probability $p > 0$. A memory-impostor is a rollout whose model score is high because retrieved cases are stale or support mismatched, but whose true return is low under the environment dynamics. Suppose every impostor has proxy score s_I and true utility u_I , and every grounded candidate has proxy score s_G and true utility u_G , with $s_I > s_G$ but $u_I < u_G$.

Proposition 1 (Finite-N memory-impostor amplification). *If a Best-of-N selector chooses the candidate with highest proxy score from N independent samples under the two-type model above, then*

$$P(\text{impostor selected}) = 1 - (1 - p)^N,$$

and the expected true utility of the selected candidate is

$$\mathbb{E}[u_N] = (1 - (1 - p)^N) u_I + (1 - p)^N u_G,$$

which decreases monotonically in N .

The proposition is not presented as deep probability theory. Its purpose is to turn a reviewer’s likely objection into a diagnostic: if memory retrieval can produce high-proxy low-true candidates, increasing N should increase proxy score, selected memory staleness, and proxy-true gap while decreasing true return.

4 DIAGNOSTIC AND REPAIR

For a selected rollout τ , define the memory exploitation gap

$$\text{MEG}(\tau) = S_{\text{model}}(\tau) - U_{\text{true}}(\tau),$$

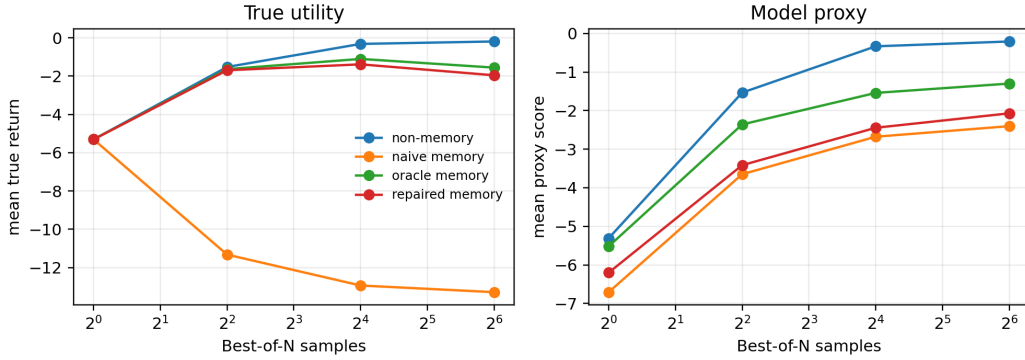


Figure 1: In the stale-memory condition, naive memory Best-of-N improves model proxy score as N grows while true return degrades. The repaired selector tracks the non-memory/oracle direction because it suppresses stale retrieved transitions.

where S_{model} is the model proxy used for selection and U_{true} is the environment return used only for evaluation. We also log retrieval precision, stale retrieval rate, disagreement among retrieved transition signs, and counterfactual memory dropout variance: the variance of predicted deltas under leave-one-retrieved-case-out perturbations.

The repair intentionally uses only observable memory metadata and transition statistics, not the hidden regime label. It has three parts. First, retrieval is recency-aware, so recent calibration transitions can compete with dense stale cases. Second, memory interpolation is gated down when retrieved cases disagree or are old. Third, the Best-of-N proxy is penalized by the average dropout-risk score. This is a small repair rather than a claim of optimality; the point is to test whether the diagnosis points toward an effective intervention.

5 CONTROLLED BENCHMARK

The benchmark is a one-dimensional latent-regime world. The observed state is position x and actions are $a \in \{-1, 0, 1\}$. The current hidden regime is *reverse*, so the true transition is $x_{t+1} = x_t - a_t$. The memory store is dense with stale *forward* transitions where $x_{t+1} = x_t + a_t$ and sparse with recent reverse calibration transitions. A naive nearest-neighbor retriever sees only (x, a) , so the stale forward cases often dominate.

Candidate action sequences are sampled with both directional modes. Each model rolls out every candidate, scores the predicted final distance to a fixed goal, and executes the proxy-best sequence under the true reverse dynamics. We compare four selectors: a non-memory model fit from recent calibration transitions, naive memory, oracle memory that is allowed to filter by the hidden regime, and the repaired memory model.

6 RESULTS

Figure 1 shows the main scaling pattern. At stale fraction 0.85, naive memory Best-of-N falls from -5.30 true return at $N = 1$ to -13.29 at $N = 64$, even though its selected proxy score is -2.40. The resulting exploitation gap is 10.88. The non-memory baseline reaches -0.19, the repaired model reaches -1.95, and the oracle retriever reaches -1.55.

Figure 2 confirms that the failure is not merely bad random actions. At $N = 64$, naive selected rollouts have 13.7% retrieval precision and 86.3% stale retrieval. Their hallucinated rollout rate is 27.8%. The repaired model improves selected retrieval precision to 47.7%, lowers stale retrieval to 52.3%, and reduces hallucinated rollouts to 0.0%.

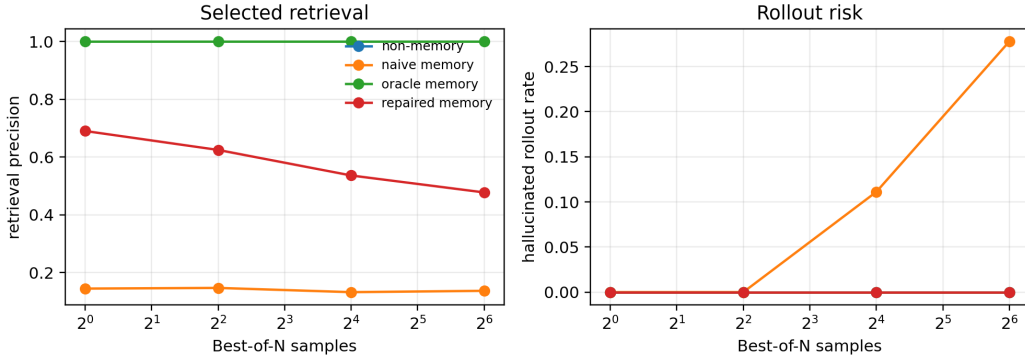


Figure 2: Selected-rollout diagnostics reveal the mechanism. Naive memory selections have poor retrieval precision and a high hallucinated-rollout rate at large N ; the repair improves selected retrieval quality and sharply reduces hallucinated rollouts.

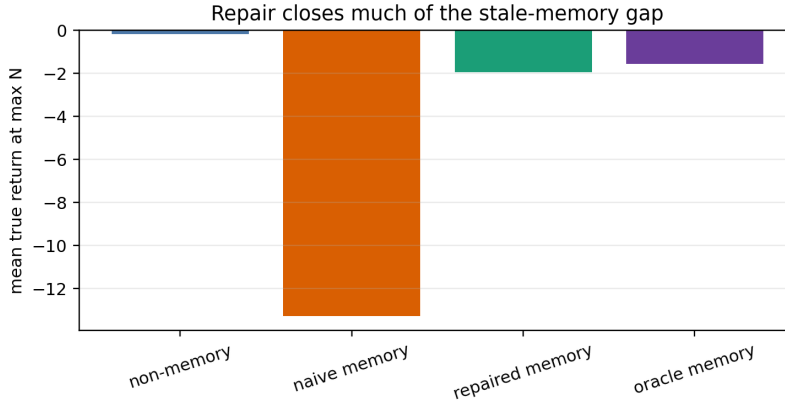


Figure 3: At the largest N , the repair improves true return by 11.33 over naive memory. The remaining oracle gap is 0.40, which is expected because the oracle can directly filter by hidden regime.

Figure 3 makes the repair boundary visible. The repaired model is not an oracle and does not claim to recover the hidden regime perfectly. It uses age, disagreement, and memory-dropout risk as support diagnostics, which are available to a memory system without revealing the environment’s latent label.

7 LIMITATIONS

The benchmark is synthetic and deliberately low-dimensional. That is a strength for mechanism isolation and a weakness for external validity. The repair is also minimal: realistic systems would need stronger retrieval calibration, diversity constraints, causal or regime-aware memories, and task-specific uncertainty models. Finally, the formal claim is only the two-type finite- N amplification law; it does not prove that a particular neural retriever will produce impostors in deployment.

8 REPRODUCIBILITY AND CHECKLIST

The repository includes the full synthetic generator, tests, raw result CSVs, plotting scripts, the 100+ entry related-work matrix, and this anonymous ICLR source. The oracle baseline is explicitly marked as using the hidden latent regime, while the repair is marked as non-oracle. No human-subject data or private datasets are used. Compute is a CPU-only synthetic run.

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A ADDITIONAL DIAGNOSTIC DEFINITION

The hallucinated-rollout indicator is one when the selected candidate’s model proxy is near-goal while its true return is poor. In code, this corresponds to a proxy score above -1.75 and true return below -5.0 . The threshold is used only for diagnostics; selection itself uses the raw proxy or repaired proxy.

B ANONYMOUS SUBMISSION NOTE

The source uses `iclr2026.conference.sty` and keeps `\iclrfinalcopy` commented. The resulting PDF displays the standard under-review anonymous author block.